

RATHAN JAYANATH SINGAVARAPU

Jersey City, New Jersey 07304 | rathan.jayanath@gmail.com | +1 5512298869 | [Linkedin](#) | [GitHub](#) | [Portfolio](#) |

SUMMARY

Detail-oriented and innovative Master's student in Computer Science with a strong foundation in software development, machine learning, and image processing. Proven experience in developing full-stack applications and implementing deep learning algorithms.

ACADEMIC PROJECTS

- BeautyU Mobile Application – React Based Mobile** [GitHub](#) November 2024 – December 2024
- BeautyU is a React-based mobile app for browsing and purchasing beauty products, featuring personalized recommendations and tailored tutorials.
 - Designed the app using React TypeScript, Bootstrap for Frontend, Firestore for Backend and Clerk authentication, implementing key features like product recommendations and user-specific tutorials.
- uSpark ML Integrated Web Application** [GitHub](#) January 2025 – Present
- A Smart Web Application that does medical diagnosis and bio medical image processing using RAG based chatbot and Unet based Medical Assistant respectively.
 - Carried out the execution of Appointment booking system for patients and workflows for both patients and doctors
 - Developed an Integrated full stack system (Front end and Back end) to utilize chatbot and Medical Assistant.
 - The application is structured with vite react framework, Redux for State management, Mongo DB and AWS for storage, Firebase and tokenization for authorization.
- Sew Smart – React Based Web** [GitHub](#) November 2024 – December 2024
- Sew-Smart is a React-based web application that allows users to upload, view, like, and comment on costume designs. It includes user authentication, a feed page, and features for viewing tailor profiles, adding designs to a cart, wish listing items, and making purchases.
 - Built the application using Vite for the frontend and Node.js with Express for the backend, integrated MongoDB for user data storage and DigitalOcean for image hosting, and used Clerk for user authentication.

WORK EXPERIENCE

- Teradata** Hyderabad,Telangana,India
Associate Software Engineer June 2022 – August 2023
- Optimized code and reduced implementation lines for database features, enhancing efficiency and stability by resolving 300+ variable allocation issues.
 - Incorporated Neo memory management strategies with few tweaks for real-time allocations. Streamlined spool and flag functions and conducted sandbox testing to ensure successful migration to database version.
- Indian Institute of Information Technology Design and Manufacturing** Chennai, Tamil Nadu, India
Research Intern August 2021– October 2021
- Implemented image processing algorithms, including U-Net, histogram segmentation, and K-means clustering, to segment ventricles in brain MRI images
 - Achieved an ORB similarity score of 0.69 with histogram segmentation, 0.11 with K-means clustering, and 95% average pixel accuracy with U-Net when compared to ground truth.
 - Classified MRI images as Alzheimer's Disease (AD), Mild Cognitive Impairment (MCI), or Normal based on segmented white pixel values, obtaining average pixel values of 4033.5 (Normal), 4062.46 (MCI), and 5088.6 (AD).
- Gustovalley Technovations** Chennai, Tamil Nadu, India
Intern March 2021 – April 2021
- Developed and hosted web applications, managed PHP requests, and executed Java-related projects to enhance system functionality, implemented hardware simulation systems and contributed to mobile application development for improved user experiences.

EDUCATION

- Pace University** New York
Masters in Computer Science - GPA: 3.75/4.0 September 2023 – May 2025
Relevant Coursework: Full Stack Enterprise App Dev, Dbms, Mobile web Content Dev, Python for Data Science
- Indian Institute of Information Technology Design and Manufacturing, Kancheepuram** Chennai,Tamil Nadu
Bachelors in Electronics and Communication Engineering - GPA: 8.1/10.0 August 2018 – May 2022

PUBLICATIONS

- Rocky Mountain Bioengineering Symposium** March 2022
"Automatic Segmentation of Endoscopic lesions using U-NET BASED DEEP ARCHITECTURE" to Biomedical Sciences Instrumentation

SKILLS

Technical: JAVA, Python, C++, HTML, CSS, JAVA script, Type Script, React Native, ML, AWS, REST Api, MERN Stack, Node.js, Express